Date:

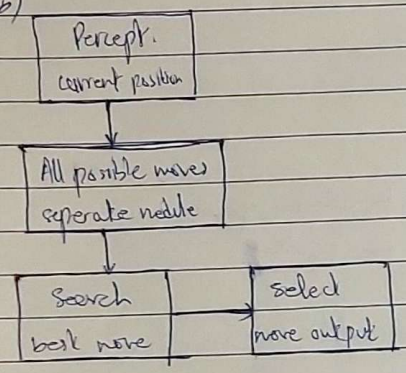
Sun Mon Tue Wed Thu Fri Sat

AI Assignment #1

Zain ul abideen Ahmad 24k0818.

Q1a) In deep blue, the objective is to best chess player using AI by using many different search algorithms to reach best possible move using knowledge and reasoning.

b)



c) It uses parallel computing with multiple chips working at a time to calculate using previous knowledge in database and search and reasoning for new.

d) It changes the chess forever, It is now the best it can be for chess only.

e) It is very narrow as only for chess and requires expensive hardware and hard coding.

f). Addition of neural network to improve self learning and learning from its mistakes.

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Q2 a) Computer Aided diagnosis uses deep learning. Image Analysis also is used to detect early tumors in X rays / MRI's

b) It use layers of filters to sort out patterns that match with medical diseases already in database.

c) Pros: faster, early detection, consistent.
Cons: Hardware Expense, AI reasoning can malfunction.

d) The improvement can be of a AI that explains its way of thinking in every part of a process. Also it will provide/give an highlighted area on the scan for us and for it to then use it for further processing.

Q3.- a) Resume.

P: Accuracy of Resume | E= Company Job Application | A = Short listing, rejecting

S= text, experience, feedback.

Flood:

P: Alerts before happening | E: River, weather | A = Send a text notify superior.

S= weather satellite, water level sensor.

b)

	Observable	Det	Ep/Seq	Stat / Dyn
Resume	Full	Det	Ep	Static
Flood	Half	Stochastic	Seq	Dynamic

CamScanner

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Q4: Goal Based:

State: (x,y) coordinates, (yes/no) Hook

Initial: No Books, start state

Action: (up, down, right, left) (pick, drop).

TM: change in coordinates according to movement (x,y).

Goal test: At target, has book

Path Cost: Total moves taken

Q5. 1) Prime uses IDDF, Iterative Deepening, DFS or IDS.

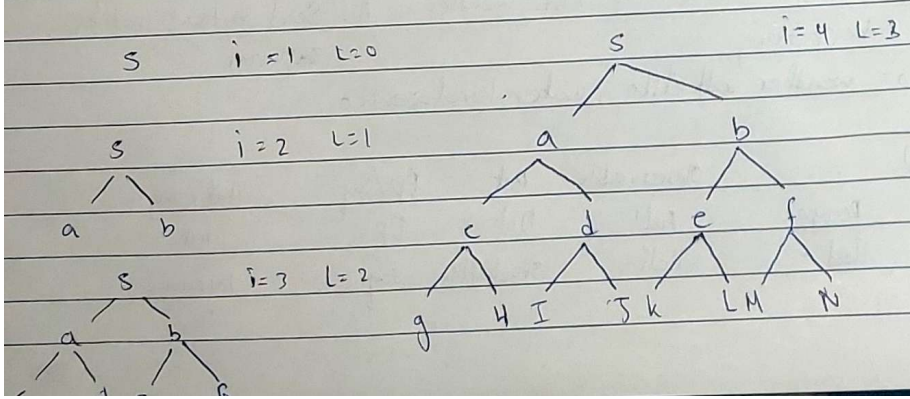
- Its method is like DFS but its Depth has limit, and the dept increment as the search go on, until the output state is not found.

2) limited memory: It stores only the current active path unlike BFS which uses more memory.

3) unknown Depth: It starts with a short limit of depth before increasing its limit

very deep dead end Tunnels: DFS can lead to dead ends but in IDDFS using limit depth this is avoided

iteration: depth: L



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Q6:

3): IDS is more suitable because

- IDS searches in levels monitoring shallow limit depth
- Less/no chance of deep end.
- less memory used due to limit depth